

Bruce Zhang

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Research Interests

Computer Science graduate with research interests in AI Systems, Autonomous Driving, and Multimodal World Models. Strong technical background in Python, Java, C++, PyTorch/TensorFlow, computer vision, deep learning, distributed systems, and performance optimization. With solid academic and technical writing skills for literature review, experimental analysis, and research documentation.

EDUCATION

Worcester Polytechnic Institute - MS in Computer Science

Courses: Operating Systems, Artificial Intelligence, Algorithms, Machine Learning, Big Data, Design of Software Systems

GPA: 3.6 Aug. 2023 - May. 2025

Dalian Institute of Science and Technology - BS in Software Engineering

Courses: Linear Algebra, Computer Networks, C Programming, Data Structures, Software Engineering, Java and Database

GPA: 2.75 Sep. 2016 - Jun. 2020

PROJECTS

BCIC(Boston Capital Investment Club)

Software Engineer - Data & Automation, MA/US

May. 2025 - May. 2026

- Built Python/C++ automation tools for log parsing, data validation, anomaly detection, and recurring reporting workflows, reducing manual review effort and improving data reliability.
- Developed data processing scripts with input validation, error handling, and integrity checks to detect inconsistent records, missing values, and abnormal changes in internal datasets.

SIR Epidemiological Model & Interactive Simulation

Jan. 2025 - Mar. 2025

(Python | FastAPI | SciPy | NumPy | React | Plotly.js | REST API)

- Developed an interactive SIR epidemiological simulation system using Python, FastAPI, SciPy, and NumPy to model disease transmission through susceptible, infected, and recovered populations.
- Implemented numerical ODE solving with SciPy to simulate population dynamics based on configurable transmission rate, recovery rate, initial conditions, and simulation duration.
- Built REST APIs connecting the numerical simulation backend with a React and Plotly.js frontend for interactive parameter adjustment and visualization.
- Visualized time-series S/I/R population dynamics with interactive plots, enabling users to explore how epidemiological parameters affect disease propagation and model behavior.

Linear Algebra Visualization Playground

Jan. 2025 - Mar. 2025

(Python | FastAPI | NumPy | React | Plotly.js | REST API)

- Developed an interactive linear algebra visualization platform using Python, FastAPI, NumPy, React, and Plotly.js to demonstrate geometric interpretations of matrix transformations.
- Implemented numerical computation for 2D vector and matrix operations, supporting configurable transformation matrices and user-defined vectors.
- Built REST APIs to perform backend numerical computations and integrated them with an interactive frontend for real-time visualization.
- Visualized vector transformations, coordinate changes, and geometric effects of linear transformations through animated plots, providing an intuitive representation of core linear algebra concepts.

MediaScout: A Personalized Twitch Resources Recommendation Platform

Jan. 2024 - Aug. 2024

(Spring Boot, React, MySQL, Spring Security, Caffeine, RabbitMQ, Jenkins, SonarQube, AWS RDS/S3, NGINX)

- Built a full-stack recommendation system (Spring Boot + React), reducing integration complexity by 40%.
- Implemented a content-based recommendation algorithm using user data, improving relevance and CTR by 20–25%.
- Optimized performance with Caffeine caching, reducing API latency by 35% and DB load by ~40%.
- Improved scalability using RabbitMQ and CI/CD (Jenkins, AWS, NGINX), enabling reliable async processing and horizontal scaling.

EmotionVision: Real-time Facial Expression Recognition System

Sep. 2023 - Jan. 2024

(Python, PyTorch/TensorFlow, OpenCV, CNN, Computer Vision, Real-time Inference)

- Built a real-time facial expression recognition system using webcam input to classify emotions such as happiness, anger, sadness, surprise, and neutral expressions.

- Trained a CNN-based deep learning model with PyTorch/TensorFlow, achieving ~85% validation accuracy through preprocessing, data augmentation, and hyperparameter tuning.
- Integrated OpenCV to capture live camera frames, detect faces, run real-time inference, and display emotion-based text feedback on screen.
- Reduced inference latency to under 100ms per frame, enabling smooth real-time emotion detection from webcam video streams.
- Visualized model accuracy, loss curves, and prediction results to evaluate performance and guide model optimization.

Ourpalm - Playcrab: Multiplayer Strategy Game

Gaming Server Engineer, Beijing/China

Jun. 2020 - Jul. 2023

Game AI & Real - Time Gameplay Systems

(C++ | A* | Dijkstra | Spatial Data Structures | 3D Vector Math | Unity/C#)

- Developed C++ pathfinding and combat AI modules using A*, Dijkstra, spatial data structures, and 3D vector mathematics for real-time navigation, target selection, and combat behaviors.
- Designed efficient spatial queries and optimized pathfinding computations for large-scale battle environments with dynamic units and obstacles.
- Integrated AI and gameplay logic with Unity/C# client systems to support real-time character movement and combat simulation.
- Optimized algorithm execution and memory usage, reducing CPU overhead by approximately 25% in large-scale battle scenarios.

Distributed Microservices & Cloud Infrastructure

(Java | Go | Spring Boot | gRPC | Kafka | RabbitMQ | Redis | MySQL | Docker | Kubernetes | AWS | Terraform | Jenkins)

- Built scalable Java/Go microservices with REST/gRPC APIs, Redis caching, MySQL, and asynchronous messaging using Kafka/RabbitMQ, supporting real-time communication for large-scale multiplayer systems.
- Improved system reliability with Consul-based service discovery, circuit breaking, fallback strategies, and fault-tolerant asynchronous processing.
- Containerized and deployed services on AWS using Docker, Kubernetes, Terraform, and Jenkins, with Deployments, Services, and HPA for scalable cloud infrastructure.
- Monitored distributed services using CloudWatch, Dynatrace, logs, metrics, and tracing to identify latency, resource bottlenecks, and cross-service failures.
- Reduced gameplay data synchronization latency by 30% and shortened release cycles by 40% through caching, payload optimization, automated deployment, and infrastructure improvements.

COMPUTER SKILLS

Languages: C++, Java, Go, Python, JavaScript

Backend: Spring Boot, Microservices, RESTful APIs, gRPC

Databases & Messaging: MySQL, Redis, Kafka, RabbitMQ

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins